



Exp :8 Write the python program to implement DFS.

Aim:

To write a Python program to implement the Depth First Search (DFS) graph traversal algorithm.

Algorithm:

Step 1: Start by defining a recursive function for DFS that takes the graph, the starting node, and a set of visited nodes as parameters

Step 2: Check if the current node is not in the visited set.

Step 3: If it is not visited, print the node and add it to the visited set to mark it as processed.

Step 4: Iterate through all the adjacent nodes (neighbors) of the current node using the graph's adjacency list.

Step 5: For each adjacent node, recursively call the DFS function.

Step 6: In the main section of the program, define a graph using a dictionary where keys are nodes and values are lists of adjacent nodes.

Step 7: Initialize an empty set to keep track of visited nodes.

Step 8 : Call the DFS function by passing the graph, the starting node, and the visited set.

Program:

```
def dfs(graph, node, visited):  
    if node not in visited:  
        print(node, end=" ")  
        visited.add(node)  
        for neighbour in graph[node]:  
            dfs(graph, neighbour, visited)
```

```
graph = {
```

```
'1': ['2', '3'],  
'2': ['4', '5'],  
'3': ['6'],  
'4': [],  
'5': ['6'],  
'6': []  
}
```

```
visited = set()  
print("DFS Traversal output:")  
dfs(graph, '1', visited)  
print()
```

OUTPUT:

```
DFS Traversal output:  
1 2 4 5 6 3  
,
```

Result: Thus, the program was executed and found that the output was found to be correct.